



Cheapest-To-Deliver Curves in MLib

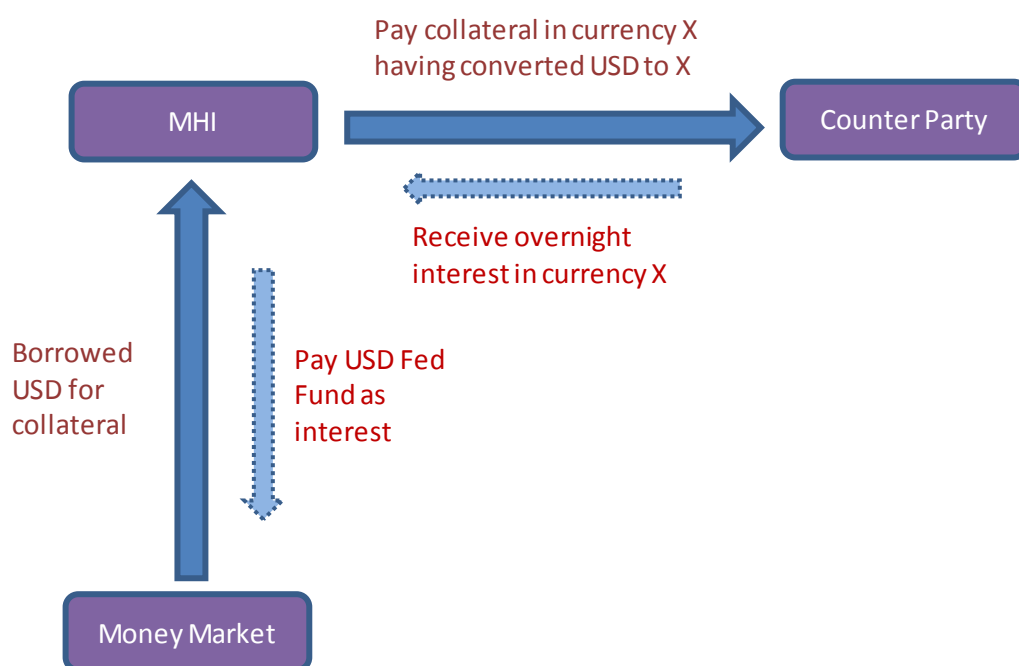
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This document serves as a simple guide on how MLib sets up its Cheapest-To-Deliver curves in a deterministic approach. It will go on to give illustrative examples on how to construct CTD curves and demonstrates how they enable practitioners to select appropriate collateral. It also provides several pricing case studies involving interest rate swaps.

What is Cheapest-To-Deliver Curve?

Credit Support Annex agreements nowadays allow counterparties to choose collateral from a big basket of eligible currencies and securities. The CTD curve is a blended collateral discounting curve, optimised through the trade life cycle among the different currency collateral curves. Product discounting based on the CTD collateral enables the best funding benefits and highest rate of return.

Imagine a USD swap is being dealt between MHI and a counterparty. Collateral is permitted in currency X by the party posting collateral which can be chosen from a basket of eligible currencies. Assume MHI is in negative MTM and needs to post collateral; the transactions typically look like this:



In this illustration MHI has borrowed the cash collateral in USD from a money market desk and the interest rate for this transaction is always in USD Fed Fund rate. Borrowed money is subsequently converted to currency X according to the CSA agreement.

It's important to note that the collateral settlement transaction in this illustration imitates a cross currency swap. However, because MHI never holds the collateral and any borrowed money is passed straight on to the Counterparty (the same with the reverse process), from MHI's point of view the notional exchanges in this imaginary cross currency swap transaction can be ignored.

The transaction has effectively simplified to:



Note that in this transaction both the forecasting and discounting are through the same curve on either side. The PV differential of the USD leg and the currency X leg (assuming notional is 1) on a certain collateral settlement date t has become:

$$\begin{aligned}
 &PV_X(t) - PV_{USD}(t) \\
 &= (1 - DF_X(t, T)) - (1 - DF_{USD}(t, T)) \\
 &= DF_{USD}(t, T) - DF_X(t, T)
 \end{aligned}$$

Where $DF_{USD}(t, T)$ and $DF_X(t, T)$ are the discount factors from the maturity date of this imaginary swap to its start date. In reality, this imaginary swap's maturity date is simply the next collateral settlement date because the CTD analysis is carried out between each two adjacent collateral dates.

From this analysis we can draw this **conclusion**: given a basket of eligible collateral currencies, the currency with the lowest $DF_X(t, T)$ gives the collateral posting party the best funding benefit. In the case of daily collateral settlement, the lowest $DF_X(t, T)$ is indicated by the highest overnight rate; if the collateral is posted weekly, the lowest $DF_X(t, T)$ should be indicated by the lowest forward rate from time t to one week from t , i.e. **the forward rate between two collateral settlement dates which are one week apart, which can also be called the forward zero rate**. The same idea should apply to other collateral settlement frequencies.

Note that this conclusion is true regardless of whether the basis between the different collateral discount curves are considered deterministic or dynamic. The actual CTD curve building mechanism is carried out deterministically and any modelling on the CSA basis in a dynamic setting is not within the scope of this paper.

Step-by-step CTD Curve Construction in MLib

Cheapest-To-Deliver curves are constructed in the following two steps In MLib:

1. Construct multiple discounting curves in different collateral currencies with respect to the same trade currency.

For example, if the trade currency is EUR and the Credit Support Annex agreement allows for the use of either EUR, USD or SEK as the collateral currency, one needs to have the following three curves constructed first:

EUR OIS curve (EUROIS)

USD OIS curve denominated in EUR (EURDF_USDCSA)

SEK OIS curve denominated in EUR (EURDF_SEKCSA)

These three curves are respectively an OIS curve, a XCCY basis curve, and a FX Forward constant curve in MLib. This paper doesn't dwell on the details of their own constructions however.

2. Pass these three curves to the MLib function, **meCurveCalibrateCTD**, which will determine the most beneficial currency in which collateral is posted on every collateral settlement day, and then subsequently build a blended CTD curve.

(a) Input:

Curve Build Properties

Curve Build Properties	
AsOfDate	21-Sep-16
Currency	EUR
Calendar	LNB:NYB:TGT
daycount	ACT/360
frequency	Daily
slidingrule	MOD_FOLLOWING
interpolation	FN_SPLINEINTERPOLATION
yieldgen.interpolation	FN_SPLINEINTERPOLATION

Collateral Curves

EURDF_USDCSA
EURDF_SEKCSA
EUROIS

(b) Curve Calibration:

Assumption:

At the beginning of the CTD curve calibration it is assumed that all underlying curves as well as the blended curve have the same zero rate convention as defined in the curve input. This is so that the forward zero rates are calculated off all the relevant collateral curves for consistent comparisons.

Algorithm:

As explained earlier, when going from time t_i to the next collateral settlement date t_{i+1} , the CTD curve would take the biggest forward zero rate:

$$F_{CTD}(t_i, t_{i+1}) = \max_{1 \leq j \leq n} (F_j(t_i, t_{i+1}))$$

Or equivalently the smallest discount factor:

$$DF_{CTD}(t_i, t_{i+1}) = \min_{1 \leq j \leq n} (DF_j(t_i, t_{i+1}))$$

Working on the discount factors in a bootstrapping approach, the discount factors of the entire blended curve are simply derived as:

$$DF_{CTD}(t_0, T) = \prod_{i=t_0}^T \min_{1 \leq j \leq n} (DF_j(t_i, t_{i+1}))$$

Where j is the index of one of the basket of n collateral curves and i is index of the collateral settlement dates.

(c) Output:

Figure 1: Showing the three independent collateral curves only

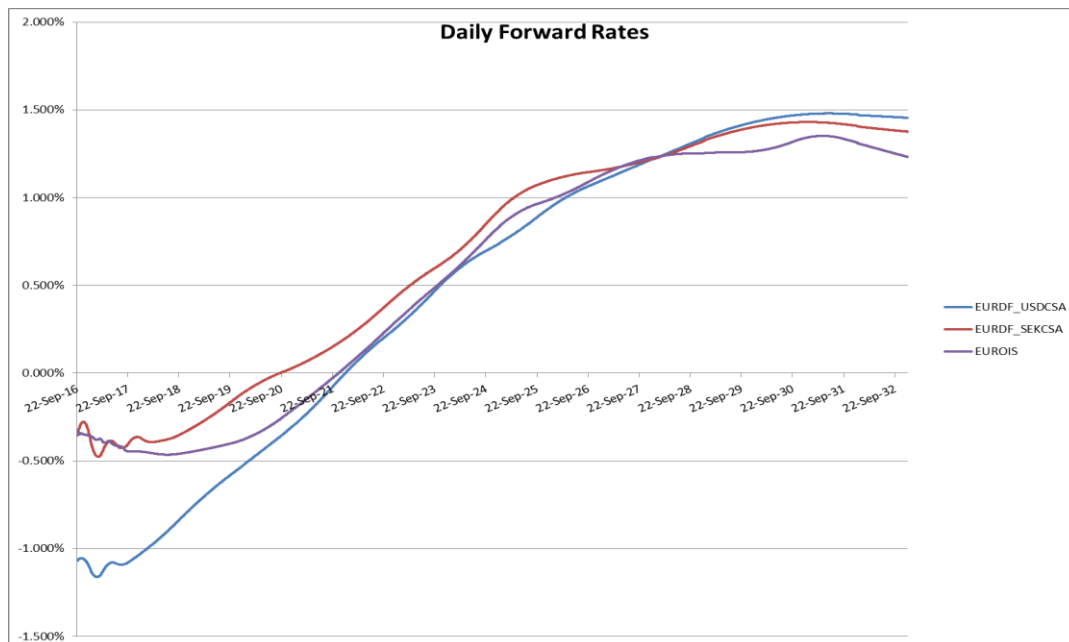


Figure 2: Showing the three independent collateral curves and the CTD blended curve

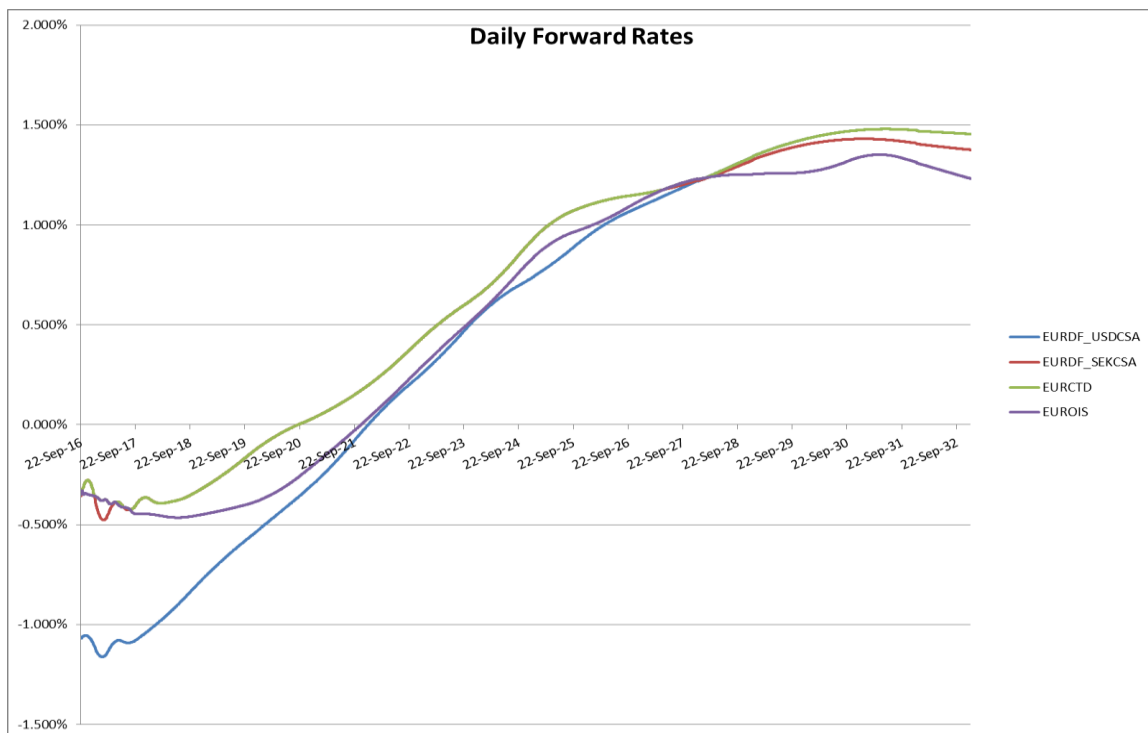
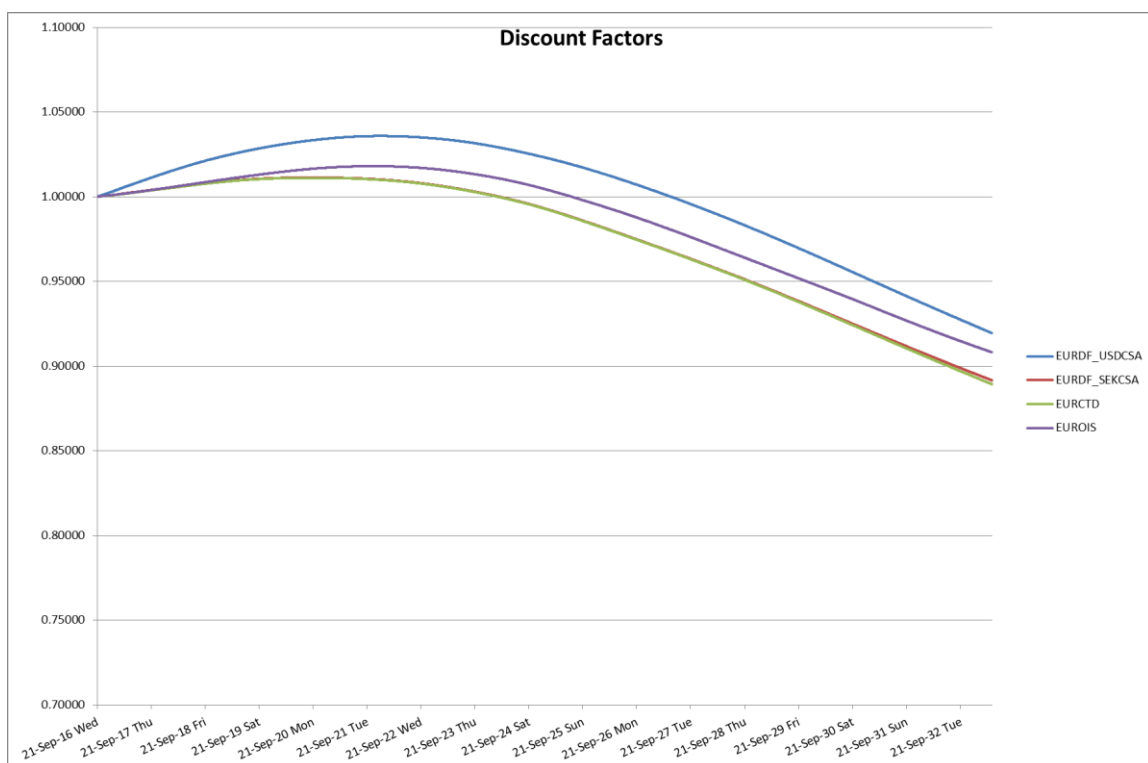


Figure 2 clearly shows that the CTD curve (green line) is closely tracking the highest overnight rate out of the three curves all the time because the collateral posting frequency is defined as Daily.

Figure 3: Showing the same Cheapest-To-Deliver result with discount curves.



We can see in Figure 3 that in the first 10 years or so the CTD discount rates are closely matching those of the SEK CSA curve. After that the CTD discount curve starts to dip further with respect to the SEK CSA curve. This is because after the first 10 years the USD CSA discount curve becomes more downward steep compared to the SEK CSA curve and the CTD mechanism has picked up on that and start behaving in the same way. This behaviour coincides with what the forward rates have presented in Figure 2.